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Trial Regeneration With Subband Signals for Motor Imagery Classification in BCI Paradigm

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ABSTRACT Electroencephalography (EEG) captures the electrical activities of human brain. It is an easy and cost effective tool to characterize motor imager (MI) task used in brain computer interface (BCI) implementation. The MI task is represented by short time trial of multichannel EEG. In this paper, the raw EEG trial is regenerated using narrowband signals obtained from individual channel. Each channel of EEG trial is decomposed into a set of subband signals using multivariate discrete wavelet transform. The selected subbands are organized in two different ways namely vertical arrangement of subbands (VaS) and horizontal arrangement of subbands (HaS) to regenerate the trials. The features are extracted from each of the arrangements using common spatial pattern (CSP). An optimum number of features are used to classify the motor imagery tasks represented by EEG trials. The effectiveness of two classifiers – linear discriminant analysis (LDA) and support vector machine (SVM) are studied. The performances of the proposed methods are evaluated using publicly available benchmark datasets. The experimental results show that it performs better than the recently developed algorithms.

INDEX TERMS Brain computer interface (BCI), discrete wavelet transformation, electroencephalography (EEG), motor imagery, narrowband signals, subband decomposition.

I. INTRODUCTION

Brain-computer interface (BCI) is a relatively new technology that helps in motor rehabilitation and neuromuscular disorder of the paralyzed patients. It is a tool which offers a communication channel and control capabilities that allow direct connection between human brain and external device using brain activities [1]. The technology used in BCI translates human intention into command which cannot pass through the brain's normal output pathway of peripheral nerves and muscles [2]. Recently, several attempts are reported to achieve the goal [3] of BCI. The electroencephalography (EEG) based BCI system is cost effective and easy to implement with minimal clinical risk as well as invasive in nature [4], [5]. It captures the brain activities as an electrical signal from the scalp using a set of electrodes.

Motor imagery (MI) is a common mental task that is widely used in BCI implementation [6]. In MI based BCI, a subject is required to perform an imagination in the brain corresponding to a specific mental task. The recorded EEG signals related to MI are classified to translate into corresponding control command for different imaginary tasks like movement of hand, foot etc. [7]. In terms of neurophysiology, motor imagery accompanies attenuation or enhancement of rhythmical synchrony over the sensorimotor cortex with the frequency bands of alpha (8 – 13 Hz) and beta (14 – 30 Hz) [8]–[11]. This paper focuses on EEG based classification of two motor imagery tasks.

The most of the current MI based BCIs have three stages including pre-processing, feature extraction and classification [12]. To improve its classification accuracy, the pre-processing is an important step in which the unwanted components of signals are suppressed. The real world multichannel EEG signal is non-stationary. The subband based approach to EEG classification performs better than that of full band method. Fourier transform (FT) based filtering is usually used for subband decomposition of EEG signal. Multiband tangent space mapping is used in [13] to extract the

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narrowband features for discrimination of MI tasks. Several studies present filter bank common spatial pattern (FBCSP) approach to enhance the performance of spatial filtering technique named common spatial pattern (CSP) to extract features used in EEG classification [14]. The CSP is one the popular feature extraction method of spatial filtering [15]. The filter-bank comprises of FT based multiple band-pass filters. It decomposes the signal into a set of subbands with predefined bandwidths. The CSP based features are extracted from each subband and combined them to implement the classification scheme. The subband CSP based feature is used in [16] to access the efficiency of narrowband signal components for MI classification. The effective features are selected using neighborhood component analysis method. A combination of brain function connectivity mechanism and one versus the rest filterbank CSP is used to improve the MI classification performance in [17]. The multi-kernel relevance vector machine (RVM) is utilized for classification. In [18], a selection scheme of CSP features is implemented considering the non-stationarity nature of EEG. The outlier suppression method is also applied to increase the interclass distances. The potential features are generated by a combination of spectral-spatial characteristics of EEG as well as the discriminative brain signal patterns [19]. The performances are studied for both subject dependent and independent cases. Frequency-optimized local region CSP (LRFCSP) is introduced in [20] to extract spatial region based features for MI classification. A few local regions are selected to derive the optimum set of features to improve the classification performance.

The EEG is captured using a set of spatially distributed electrodes. All the electrodes are not equally important to MI classification. Hence, the selection of effective channels (electrodes) has recently taken into consideration. A correlation-based channel selection (CCS) method is introduced in [21] to select the channels that contain more correlated information. A regularized CSP (RCSP) method is used to extract effective features to perform the MI classification. The genetic algorithm (GA) is also used for channel selection to extract the most relevant channels for enhanced classification [22]. The distinctive channels in terms of correlation coefficient (CC) values are selected to enhance the classification performance [23]. The features based on filter-bank CSP (FBCSP) is used for MI classification using SVM. The highly correlated channels are selected based on the highest Fisher ratio of time domain parameters together with correlation coefficients (TDP-CC) [24]. The bi-spectrum based features are used for channel selection in [25]. Thus selection of effective channels improves the performance of MI classification. The selection of proper time-window is another issue for MI classification. The correlation-based time window selection (CTWS) approach is used to adjust the starting point of EEG with motor imagery (MI) [26]. The proper selection of time window leads to enhance the performance of MI classification implemented by SVM.

Empirical mode decomposition (EMD) has been applied to EEG analysis in BCI application to mitigate the problem of non-stationarity [27], [28]. It is fully data adaptive and suitable to analyze nonlinear and nonstationary signals. Due to the lack of proper mathematical model, EMD has non-consistency and mode mixing problem in analyzing multichannel EEG [29]. The multivariate extension of EMD (MEMD) has been applied for EEG signal decomposition to resolve the problems. It ensures the proper localization of frequency and amplitude [30]. The mode mixing problem is completely resolved by introducing noise-assisted MEMD (NAMEMD) [31]. The NAMEMD-CSP based approach is used to improve classification accuracy of MI tasks by enhancement of frequency component localization of EEG signals [11]. It shows better performance than conventional methods including short term Fourier transform [32], wavelet transform and synchrosqueezed wavelet transform [33]. All the EMD based approaches require high computational complexity. The required computation cost is one of obstacles for real time BCI implementation. To resolve the problem, discrete wavelet transform (DWT) is widely used for signal decomposition into a finite set of subbands [34]. The multivariate wavelet transform (mWT) is introduced in [35] to decompose multichannel EEG signal followed by feature extraction. The extracted features are applied to the machine learning system mostly implemented by linear discriminant analysis (LDA), support vector machine (SVM) and artificial neural network (ANN) [36], [37].

In this paper, the features extracted from subband signals of multichannel EEG are used in MI classification. Instead of using the original signals of different channels, the subband signals are arranged for feature extraction. The mWT is applied to implement data adaptive subband decomposition of EEG signals. The obtained subband signals are arranged in two ways – vertical and horizontal concatenation to regenerate the trials of EEG. The CSP based features are extracted from the regenerated trials. It provides the features from narrowband signals and hence more discriminative in nature. The classification is performed by using LDA and SVM.

The rest of the paper is organized as follows. Section II provides the description of data used in this study. The methodology is described in Section III. The experimental results are presented in Section IV and discussion is illustrated in Section V. Finally, Section VI draws a conclusion about this study.

II. DATA DESCRIPTION

The performances of proposed methods are evaluated using two datasets as described in the following subsections.

Dataset I: Publicly available EEG data obtained from BCI Competition IV dataset 1 [38] is used in this study for motor imagery classification. The EEG data was collected from seven healthy subjects (labeled as ‘a’, ‘b’, ‘c’, ‘d’ ‘e’, ‘f’, and ‘g’). The motor imagery was performed without feedback at the time of recording. Each subject performed two motor imagery tasks among three of left hand movement, right hand movement, and foot movement. Every different MI task is treated as a different class. Visual cues were displayed for the duration of 4s and each subject was required to perform the predefined motor imagery task. The number of total trials is 200 and each trail was randomized so that all subjects imagined two tasks equally. Trails were interleaved with 2s of a blank screen and 2s with a fixation cross shown in the center of the screen with superimposed on the cues. The EEG signal was recorded using 59 electrodes according to the international 10-20 system using BrainAmp MR plus amplifiers and an Ag/AgCl electrode cap. Signals were band-pass filtered between 0.05 and 200Hz and then digitized at 1000Hz with 16-bit resolution. In this study, the signals are down-sampled at 100Hz.

It is well known that the center part of the brain is mostly active during motor imagery [39]. We have selected 23 channels out of 59 to be used in this study. The selected channels are: "FC5", "FC3", "FC1", "FC_Z", "FC2", "FC4", "FC6", "C_Z", "C3", "C4", "C1", "C2", "C5", "C6", "T7", "T8", "CCP3", "CCP4z", "CP5", "CP1", "CP_Z", "CP2", "CP6" according to the 10-20 system [40]. It is also known that the time segment selection of motor imagery is a critical issue for EEG classification [16]. The EEG of 3.5s (0.5s to 4s) segment is used here [41]. A zero phase Butterworth bandpass (8 – 30Hz) filter is applied to all the trials before any further processing.

Dataset II: An open EEG dataset is presented for general-purpose BCI research [19], [42]. The EEG signals were recorded with a large number of subjects (54 participants) and in multiple sessions (two sessions on different days) for motor imagery (MI) tasks. The signals were digitized with sampling rate of 1000 Hz. In this study, the recorded signals are downsampled to 100 Hz. The MI paradigm was designed by following a well-established system protocol. For all blocks, the first 3 s of each trial began with a black fixation cross that appeared at the center of the monitor to prepare subjects for the MI task. Afterwards, the subject performed the imagery task of grasping with the appropriate hand for 4 s when the right or left arrow appeared as a visual cue. At the end of each task, the screen remained blank for 6 s. The experiment consisted of training and test phases; each phase had 100 trials with balanced right and left hand imagery tasks.

III. METHODS

The proposed method uses subband based features of multichannel EEG signals for MI classification leading to BCI implementation. It has four steps: (i) subband decomposition of multichannel EEG signals is performed using multivariate wavelet transform (mWT), (ii) the effective subbands are arranged in two ways (horizontal and vertical) to regenerate the trials, (iii) the features are extracted from new trials using CSP and (iv) machine learning approach is applied to classify MI tasks using the extracted features. Each step is explained in the following subsections. The block diagram of the proposed method is illustrated in Fig.1.

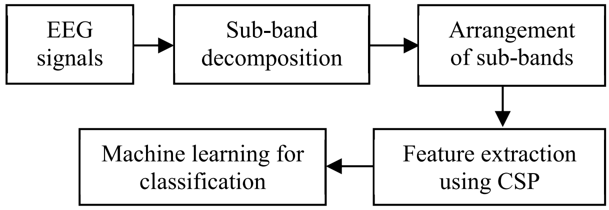

FIGURE 1. Block diagram of the proposed method for MI classification in BCIs paradigm.

A. MULTIBAND DECOMPOSITION

Discrete wavelet transform (DWT) is one of a powerful signal-processing tool for multi-resolution analysis. The empirical mode decompose allows users to conduct a data-driven decomposition for nonlinear and non-stationary signals [43]. The DWT decomposes the signal at a different resolution with different translations in time by band-pass filtering using versions scaling and wavelet functions [14]. The detail mathematical elaboration of DWT can be found in [44].

Because of significant spatial correlation of the wavelet coefficients at each decomposition level, a spatial weighted average of the decomposition structure is proposed in [35], [45] which leads to implement multivariate WT (mWT). We consider that X is a data matrix with dimensions $N \times P$, where N is the number of channels and P is the number of samples per channel with an assumption $P \gg N$. The weights are estimated at the jth level by performing an Eigen analysis of the array wavelet covariance matrix

$$w_j = \max_{\sigma}\left(E[W_j W_j^T]\right) \tag{1}$$

where W_j denotes the $N \times P/2^j$ array wavelet transform matrix at the jth level obtained by applying the $P \times P/2^j$ orthonormal wavelet transformation matrix W to the snapshot of the N channels array data matrix $X = [x^{(1)}, x^{(2)}, \ldots, x^{(N)}]$ [35]. The Eigen vector is chosen corresponding to the maximum Eigen value obtained from Eigen decomposition of array wavelet covariance matrix.

The wavelet decomposition of multichannel EEG signal X at level J produces $J+1$ matrices $\delta_1, \delta_2, \ldots, \delta_J$ containing the detail coefficients at levels 1 to J respectively and the approximation coefficients α_J of N signals. Matrices δ_j and α_J both are of size $N \times 2^{-j}P$. Subband signals are reconstructed from the detail and approximate coefficients denoted as $\delta_1, \delta_2, \ldots, \delta_J$ and α_J respectively. After mWT decomposition, the signal of the nth channel can be represented as [45]:

$$x^{(n)} = \sum\nolimits_{j=1}^{J} x_j^{(n)} + x_{J+1}^{(n)} \tag{2}$$

where $x_j^{(n)}$ is the jth subband corresponding to the detail coefficient $\delta_j^{(n)}$ at the jth level and $x_{J+1}^{(n)}$ is the $(J+1)$th subband reconstructed from the approximate coefficient $\alpha_J^{(n)}$ of the nth channel. The mWT decomposes all the signals with the same number of subbands for each channel considering the spatiotemporal correlations among the channels.

B. TRIAL REFORMATION WITH SUBBANDS

Traditionally, the spatial filtering based feature extraction is performed directly from the multichannel EEG signals. The narrowband signals contain more discriminative features for MI classification. The mWT is applied here for the multiband decomposition of EEG data $X = [x^{(1)}, x^{(2)}, \ldots, x^{(N)}]$, where $x^{(n)}$ represent nth channel of EEG trial X. The subbands obtained from nth channel is denoted by $x_j^{(n)}$, where $n = 1, 2, \ldots, N$ represent the channel indices and $j = 1, 2, \ldots, (J+1)$ is index of subband. For instance, the 1st channel $x^{(1)}$ is decomposed by mWT with J level yielding the sub-bands denoted by $x_1^{(1)}$, $x_2^{(1)}, \ldots, x_J^{(1)}+1$. The sampling frequency of EEG data used in this study is 100 Hz. The frequency ranges of the first four subband obtained by mWT with $J = 3$ are (25 – 50 Hz), (12.5 – 25 Hz), (6.25 – 12.5 Hz) and (0 – 6.25 Hz). The first three subbands are used here to access the frequency range (8–30 Hz) containing motor imagery task. The last subband (0–6.25Hz) does not contain the signal of frequency range 8–30 Hz and hence it is discarded. Thus the value of J is selected as 3.

A selected number of sub-bands are suppressed to enhance the classification performance [46], [47]. On the other hand, it is very hard to select such sub-bands after multiband decomposition. Considering the situation, all the subbands are kept and arranged in different ways to reform the trials. The narrow band signals are used to produce more discriminative features for EEG classification. The signal of each channel is decomposed into $J+1$ subbands ($J = 3$) using mWT. The obtained subbands are spatially arranged to facilitate the feature extraction method using spatial filter. The $(J+1)^{\text{th}}$ subband containing lowest frequency signal components is discarded. The spatial arrangement of the subbands is performed in two ways as described in the following subsections.

1) VERTICAL ARRANGEMENT OF SUBBANDS (VaS)

A new data matrix is derived by arranging the subbands obtained from individual channel of EEG data using mWT based decomposition. We can consider that the nth channel of EEG is decomposed into the subbands as $x_1^{(n)}, x_2^{(n)} \ldots x_j^{(n)} \ldots x_{J+1}^{(n)}$; where $x_j^{(n)}$ is the jth subband of nth channel of EEG data. Each of $x_j^{(n)}$ is a row vector. The last subband $x_{J+1}^{(n)}$ is not used in trial regeneration. Each subband is placed as a row to form new trial such that the subbands with similar frequencies are placed adjacently.

In vertical arrangement, the subbands are arranged in the following fashion to reform the trial

$$X^V = \left[\left[x_1^{(1)}\right]^T \left[x_1^{(2)}\right]^T \ldots \left[x_1^{(N)}\right]^T \left[x_2^{(1)}\right]^T \left[x_2^{(2)}\right]^T \ldots \left[x_2^{(N)}\right]^T \ldots \left[x_J^{(1)}\right]^T \left[x_J^{(2)}\right]^T \ldots \left[x_J^{(N)}\right]^T\right]^T \tag{3}$$

In this case, the first subbands (highest frequency components) of all channels are placed as individual row of lower index then the second subbands of all channels (in sequence) are placed as individual row and so on to reform the trial of size $(N \times J) \times P$. The vertical arrangement of the subbands obtained from all the channels is illustrated in Fig. 2. In the newly generated trial, each row represents a narrow band signal than the original channel of EEG signals hence better representation of discriminative features is achieved.

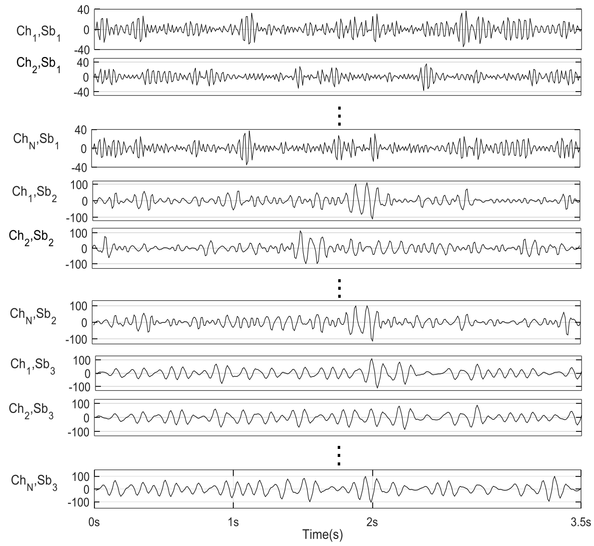

FIGURE 2. Vertical arrangement of subband signals to regenerate the trial.

2) HORIZONTAL ARRANGEMENT OF SUBBANDS (HaS)

The individual channels of EEG data are decomposed into the finite set of subbands using mWT in the similar way as described in the vertical arrangement method. In this method, the subbands obtained from individual channel are concatenated horizontally to form a new channel with different frequencies. Thus, newly formed signal of any channel contains the subband signals in different time spans. It incorporates the effect of narrowband EEG signals on motor imagery activities. The reformed trial with horizontal concatenation with subband signals can be represented by the following data matrix.

$$X^H = \begin{bmatrix} x_1^{(1)} x_2^{(1)} \ldots x_J^{(1)} \\ x_1^{(2)} x_2^{(2)} \ldots x_J^{(2)} \\ \ldots \\ x_1^{(N)} x_2^{(N)} \ldots x_J^{(N)} \end{bmatrix} \tag{4}$$

where the row vector $x_j^{(n)}$ is the jth subband obtained from nth channel of EEG signal. It is noted that the last subband $x_{J+1}^{(n)}$ is not used in trail regeneration. The trial X^h has the dimension of N rows with $(J \times P)$ columns. The horizontal arrangement of the subbands obtained from a single trial is shown in Fig. 3.

C. FEATURE EXTRACTION

Common spatial pattern (CSP) is an effective feature extraction method that has great effect on the classification of motor imagery represented by EEG. Assume that E_q^i is the multichannel EEG for trial i of class q where, E_q^i is represented by $E_q^i \in \Re^N \times P$; N is the number of channels and P is the

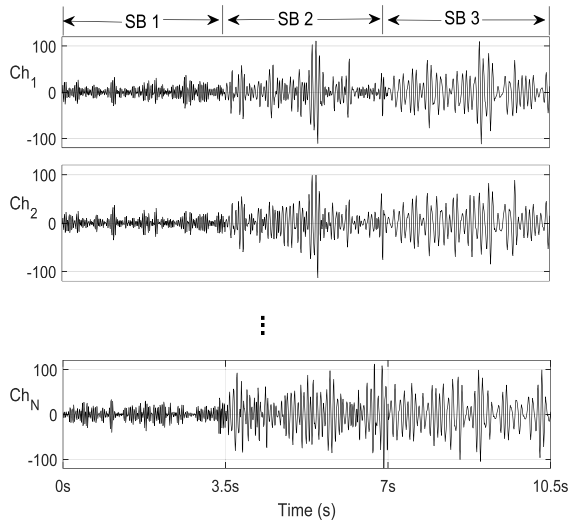

FIGURE 3. Horizontal arrangement of subband signals to regenerate the trial.

number of samples per channel. For the explanation, the EEG data of a single trial ($i = 1$) is represented as $E_{q\in(h\times f)}$ where h and f denote hand and foot movement respectively. The normalized spatial covariance of the EEG for hand movement can be calculated as:

$$C_h = \frac{E_h E_h^T}{tr(E_h E_h^T)}, \tag{5}$$

where E_h represents the original EEG matrix for hand movement, $(.)^T$ is the transpose operator and $tr(.)$ represents the sum of the diagonal elements of any given matrix [15]. Similarly, the covariance for foot movement denoted by C_f can be computed with corresponding EEG data E_f. Each covariance matrix is normalized using the trace to account for variability in the amplitude of any trial. The averaged normalized spatial covariance (of individual class) $\overline{C}_h$ and $\overline{C}_f$ are estimated by averaging over all the trials of respective class. The composite spatial covariance, C is calculated as

$$C = \overline{C}_h + \overline{C}_f = U \sum U^T \tag{6}$$

where U is the matrix of eigenvectors and Σ is the diagonal matrix containing the eigenvalues. The whitening transformation matrix P is then computed as: $P = \sum^{-\frac{1}{2}} U^T$.

The transformation matrix P equalizes the variance in the space spanned by the eigenvectors in U. P is then applied on the average covariance matrices $\overline{C}_h$ and $\overline{C}_f$ as

$$S_h = P\overline{C}_h P^T, S_f = P\overline{C}_f P^T \tag{7}$$

The transformed terms S_h and S_f share common eigenvectors and the sum of corresponding eigenvalues for the two matrices will always be 1. In other words,

$$S_h = V \sum\nolimits_h V^T, S_f = V \sum\nolimits_f V^T, \sum\nolimits_h + \sum\nolimits_f = 1 \tag{8}$$

The projection matrix W is obtained by $W = V^{\mathrm{T}} P$.

When the eigenvalues of S_h are sorted in descending, top m eigenvectors account for the maximum variance for the right hand movement (first class) and at the same time they also account minimal variance in right foot movement (second class) due to the constraint of Eq. (8). On the other hand, last m eigenvectors account minimum variance in right hand movement (first class) but maximal variance in the right foot movement (second class). Hence, first m and last m columns of W are kept in order to retain discriminative patterns. A new projection matrix W_{2m} is formed. The EEG signal is converted as:

$$Z = W_{2m}^T \frac{E}{\sqrt{tr(EE^T)}} \tag{9}$$

where Z contains the new signal in projected space. It contains common and specific components in different motor imagery tasks. For i^{th} trial, the feature vector of dimension $2m$ is computed and it contains the variance of the rows of Z as $x_i = (g_1, g_2, \ldots g_{2m})$ and g can be computed as $g_r = log(var(z_r))$, where r denotes the row number of Z and z_r represents the corresponding row vector. The features are used to train the classifier for recognition of motor imagery tasks. The features of test set are extracted by taking the projection of test trails on the spatial filter derived from the training trials of EEG.

D. CLASSIFICATION

Effective feature extraction turns the key part of BCI into pattern classification. The feature vector extracted from any EEG trial represents the respective pattern. The classification is the problem of identifying class any given observation based on the training data with available class labels. Linear discriminate analysis (LDA) and support vector machine (SVM) have been frequently adopted with CSP features in recent researches for MI classification [48], [49]. Both the classifiers are employed here to discriminate the unknown testing set of observation based on the training set of known observation using feature vector. The effectiveness of each classifier is studied with different sets of features.

1) LINEAR DISCRIMINATE ANALYSIS LDA

It is a statistical technique used for EEG data classification. The goal of LDA is to build a predictive model for a group membership [50], [51]. It is accomplished by using hyper-planes to separate data representing different classes [49]. The predictive model is composed of a linear discriminate function which maximizes the ratio between class variances in the data set [52]. Using the ration of variances, LDA efficiently classify a two class data.

2) SUPPORT VECTOR MACHINE SVM

It constructs an optimal hyperplane by maximizing the separation margin between two classes. Given training samples $(x_i, y_i), i = 1, 2, \ldots, S$; where x_i is the ith input feature vector, and y_i is the class label of x_i. The SVM requires the solution of the following optimization problem defined as [52], [53]

$$\begin{aligned} &\min_{u,b} \tfrac{1}{2} \|u\|_2^2 + C \sum_{i=1}^{S} \xi_i \\ &s.t. \quad y_i \left(u^T x_i\right) + b \geq 1 - \xi_i, \qquad \xi_i \geq 0 \end{aligned} \tag{10}$$

where u is the learned discriminate vector, C is a regularization parameter, ξ is a slack variable and b denotes a scalar. It is a constrained optimization problem and can be solved by a Lagrangian representation [54]. However, the solution uses a kernel function, $k(x_i, x_j)$ to accomplish nonlinear decision boundary for EEG classification by mapping features into higher-dimensional space. The detailed derivation and additional information are available in [38]. Different kernel functions are used here to study the performances of SVM as classifier.

IV. EXPERIMENTAL RESULTS

The experiments are conducted with Dataset I and Dataset 2 (explained in Section II) to test the efficiency of the proposed method for motor imagery classification. Binary classification is considered here to discriminate the motor imagery tasks for hand and foot movement over the entire dataset. After pre-processing, each channel of any EEG trial is decomposed into four subbands using 3 levels of multivariate wavelet transform. The lowest frequency subband (0–6.25Hz) is discarded and the rest three subbands (25–50Hz, 12.5–25Hz, 6.25–12.5Hz) are arranged in two ways (horizontal and vertical) to regenerate each trial. In vertical arrangement of subbands (VaS), the subbands are placed as individual row group by the index of subbands (subbands of same indices are placed consequently) to regenerate the trial of size $3N \times P$ rows, where N is the number of channels and P is the channel length. In horizontal arrangement of subbands (HaS), the subbands obtained from any channel are horizontally concatenated to regenerate the trial of size $N \times 3P$. The features of each arrangement are extracted using CSP. The classifier is trained by the features obtained from training data leading to evaluate the performance of the proposed method using test dataset. Two classifiers SVM and LDA are employed here to evaluate the performance of MI classification.

The performances of the proposed approaches are evaluated with k ($k = 5$) fold cross-validation. To arrange the cross validation, the dataset is randomly divided into 5 equal subsets where one of the subsets is used for the test while rests 4 are used for training. The cross validation is repeated by 5 times and then the results are averaged to yield the classification accuracy.

In this study, the performances of two different trial regeneration approaches named as VaS and HaS are evaluated with two classifiers – SVM and LDA. The performance is evaluated by the terms precision, recall and accuracy (in %) defined as

$$Precision = \frac{T_P}{T_P + F_P} \times 100 \tag{11}$$

$$Recall = \frac{T_P}{T_P + F_N} \times 100 \tag{12}$$

$$Accuracy = \frac{T_P + T_N}{T_P + F_N + T_N + F_P} \times 100 \tag{13}$$

where $\mathrm{T_P}$, $\mathrm{F_N}$, $\mathrm{T_N}$ and $\mathrm{F_P}$ represent true positive, false negative, true negative and false positive respectively.

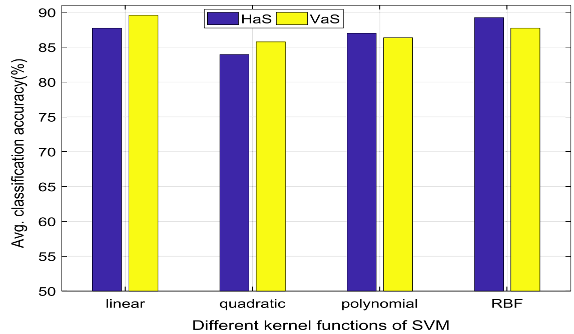

FIGURE 4. Comparison of performance (in term of classification accuracy) of VaS and HaS for Dataset I using SVM with different kernel functions.

The average classification accuracies (over the seven subjects of Dataset I) of SVM with different kernel functions for VaS and HaS are illustrated in Fig. 4. It is observed that the maximum average accuracies with VaS and HaS methods are achieved by SVM classifiers using linear kernel and radial basis function (RBF) kernel respectively. Note that the highest average accuracy of VaS is achieved by SVM with linear kernel (VaS–SVM$_{lk}$).

The performance of LDA is also evaluated for VaS and HaS approaches. The accuracy of SVM (with linear kernel) for VaS is compared with that of LDA for VaS and HaS as shown in Table 1. It is found that the highest average classification accuracy over seven subjects of Dataset I is achieved by VaS–LDA (89.84%) method although the performance of VaS–SVM$_{lk}$ (89.60%) is very close to the highest one. Besides the accuracy, the precision and recall are also illustrated to investigate the performance of the classifiers. The receiver operating characteristic (ROC) curves of three methods HaS-LDA, VaS-LDA and VaS-SVM$_{lk}$ over all of seven subjects are illustrated in Fig. 5. It is observed that the higher TPR with respect to FPR is achieved by VaS-LDA among these three methods. It reflects the performance of VaS-LDA in terms of precision, recall and accuracy. The highest average precision (90.35%) and recall (89.83%) are achieved by using VaS-LDA method. It is observed from Fig. 4 that the maximum accuracy is obtained by VaS–SVM$_{lk}$ over HaS–SVM with any kernel function. The HaS approach with LDA classifier (HaS-LDA) attains lower accuracy than that of VaS with LDA (VaS-LDA) as illustrated in Table 1. On the basis of the results presented in Fig. 4 and Table 1, it is noted that there is no significant difference between the performances of VaS and HaS approaches. It is noted that the performance (accuracy) of HaS-LDA is lower than both of VaS-LDA and VaS-SVM$_{lk}$ methods (illustrated in Table 1). The performance of the proposed method is also evaluated with Dataset II (described in Section II). The experimental setup is similar to the method used for Dataset I. The results with first five subjects are illustrated in Table 2.

The highest classification accuracy (86.10%) for first five subjects taken from Dataset II is achieved with VaS-LDA method. The performance of VaS-SVM$_{lk}$ is 0.88% lower than that of VaS-LDA while the classification accuracy of

TABLE 1. The comparison of performances (in terms of precision, recall and classification accuracy) of SVM and LDA classifiers with different trial regeneration approaches for dataset I.

Subject	HaS-LDA			VaS-LDA			VaS-SVM$_{lk}$		
	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy
a	87.01	88.40	87.60	90.00	86.40	88.40	92.25	92.80	**92.50**
b	69.04	77.60	71.40	71.55	85.00	75.60	77.00	77.00	**77.00**
c	89.32	83.60	**86.80**	89.73	80.40	85.60	82.50	83.00	82.70
d	94.77	97.80	96.20	96.67	98.60	**97.60**	96.22	96.60	96.40
e	98.81	99.60	**99.20**	98.02	99.00	98.50	97.77	96.60	97.20
f	90.35	88.00	89.30	92.32	89.00	**90.80**	88.65	89.00	88.80
g	91.95	86.80	89.60	94.17	90.40	92.40	94.19	90.80	**92.60**
Avg±Std	88.75±9.51	88.83±7.69	88.59±8.87	90.35±8.86	89.83±6.90	**89.84**±7.80	89.80±7.62	89.40±**7.21**	89.60±7.40

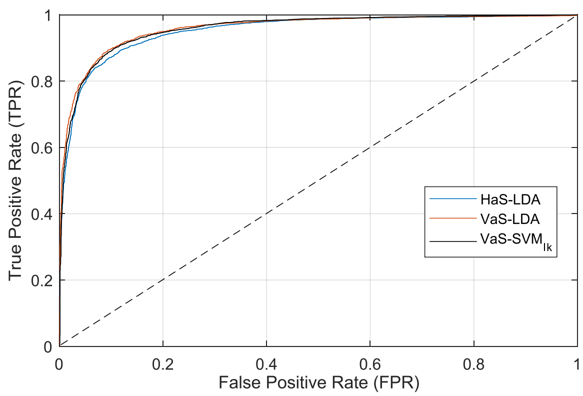

FIGURE 5. The ROC curves of three methods HaS-LDA, VaS-LDA and VaS-SVM$_{lk}$ over the seven subjects of dataset I.

TABLE 2. Performance (accuracy) of the proposed methods with the first five subjects of dataset II.

Subject	CSP-SVM	HaS-LDA	VaS-LDA	VaS-SVM$_{lk}$
s1	**78.95**	77.55	78.75	75.25
s2	92.45	94.25	**99.35**	98.65
s3	90.35	95.50	96.70	**96.75**
s4	46.80	56.90	**59.30**	59.00
s5	94.50	**96.50**	96.40	96.45
Avg±SD	80.61±17.74	84.14±**15.29**	**86.10**±15.30	85.22±15.66

HaS-LDA is 1.96% lower than the accuracy of VaS-LDA. The method CSP-SVM is implemented using CSP based features and SVM with linear kernel without trial regeneration technique. It has the lowest performance. Hence, the use of proposed subband based trial regeneration method significantly improves the performance of MI classification.

V. DISCUSSION

The MI classification performances of the proposed methods are compared with a number of recently reported methods in which the mentioned datasets are used for evaluation. Only four (a, b, f, g) subjects out of seven in Dataset I are used to evaluate the performances in several works [11], [21], [23], [24]. The comparative performances with these four subjects in terms of classification accuracy are illustrated in Table 3.

The maximum average classification accuracy (87.72%) over the mentioned four subjects is achieved using the VaS-SVM$_{lk}$. It is higher than any other methods reported in Table 3. It is noted that the highest average accuracy (89.84%) over the seven subjects of the Dataset I is attained by VaS-LDA, whereas, the performance of VaS-SVM$_{lk}$ is superior to VaS-LDA method over four subjects (a, b, f, g). The performances of proposed methods VaS-LDA as well as VaS-SVM$_{lk}$ are compared with noise assisted multivariate EMD (NA-MEMD) [11], correlation-based channel selection with regularized CSP features (CCS-RCSP) [21], channel selection using correlation coefficient with features extraction by filter-bank CSP (CC-FBCSP) [23] and channel selection time domain parameters and correlation coefficient (TDP-CC) [24].

TABLE 3. Comparison of MI classification accuracy (%) of the proposed work with recently developed methods using four subjects (a, b, f, g) of dataset I.

Method \ Subject	a	b	f	g	Avg±Std
CSP-SVM	88.70	73.50	87.90	92.20	85.58±7.15
VaS-LDA	88.40	75.60	90.80	92.40	86.80±7.64
VaS-SVM$_{lk}$	**92.50**	77.00	88.80	92.60	**87.72**±7.36
NA-MEMD [11]	85.90	77.60	78.80	90.90	83.30±6.25
CCS-RCSP [21]	85.50	67.00	79.50	**94.50**	81.60±11.50
CC-FBCSP [23]	86.50	69.50	87.50	94.00	84.40±8.20
TDP-CC [24]	86.50	57.25	**92.50**	90.50	81.69±16.80
NS-CSP [18]	82.00	67.50	65.00	87.50	75.50±11.00
BCS-CSP [25]	79.00	**79.00**	92.00	88.00	84.50±**5.69**

The non-stationary property is considered to calculate CSP (NS-CSP) in [18] and bi-spectrum based channel selection is implemented in [25]. Park *et al.* [11] has proposed MI classification algorithms using various decomposition methods such as Butterworth filter (BF), continuous wavelet transformation (CWT), synchrosquizing transformation (SST), EMD, ensemble EMD (EEMD), multivariate EMD (MEMD), and noise assisted MEMD (NA-MEMD). The alpha and beta rhythms have been used to extract the features. The maximum classification accuracy (83.30%) has been achieved for four subjects by using NA-MEDM based method in [11]. Its accuracy is 4.42% lower than that of the proposed VaS-SVM$_{lk}$. With NA-MEMD, the IMFs effective for MI classification are selected heuristically. In contrary, all the subbands are used in this study and the performance is improved.

The channel selection based MI classification methods recently draw attention of the related research community. The performances of proposed methods are compared with three channel selection based algorithms described in [21], [23], [24] as illustrated in Table 3. The effective EEG channels are selected using correlation based method, correlation coefficient, time domain parameter with correlation coefficient in CCS-RCSP [21], CC-FBCSP [23] and TDP-CC [24] respectively. The CSP based features have been used in all of the three methods. The average classification accuracy (over the four subjects of Dataset I) of the proposed VaS-SVM$_{lk}$ method is at least 3.32% higher than any one of the channel selection based approaches. The proposed methods emphasizes on the extraction of potential features from trials which are regenerated using narrowband signals of each channel. The components representing MI task are localized in frequency scale and hence the performance of MI classification is increased. The channel selection based methods [21], [23], [24] have only focused on the selection of effective channels rather than the extraction of distinctive features.

Statistical test is performed to measure the significance of the proposed method. Tukey-Kramer based post-hoc test [45] suggests that the proposed VaS-SVM$_{lk}$ method achieved significantly higher accuracy across 4 subjects for motor imagery classification than the other methods (VaS-SVM$_{lk}$ vs CSP-SVM: $p<0.05$, VaS-SVM$_{lk}$ vs VaS-LDA: $p>0.09$; VaS-SVM$_{lk}$ vs NA-MEMD: $p>0.07$, VaS-SVM$_{lk}$ vs CCS-RCSP: $p<0.05$, VaS-SVM$_{lk}$ vs CC-FBCSP: $p>0.07$, VaS-SVM$_{lk}$ vs TDP-CC: $p<0.05$, VaS-SVM$_{lk}$ vs NS-CSP: $p<0.03$, VaS-SVM$_{lk}$ vs BCS-CSP: $p>0.15$).

The performances of different classifiers as well as two proposed methods are evaluated using all of seven subjects of Dataset I as shown in Table 1. The frequency-optimized spatial region based CSP features are introduced in (LRFCSP) [20]. The effective features are obtained from the selected spatial regions. Feng *et al.* [26 proposed a novel correlation-based time window selection (CTWS) algorithm to solve the problem of fixed time window of the MI-based BCIs system. The few local regions are selected to derive the optimum set of features to improve the classification performance. The subspace optimization based feature has been used to attain optimal performance. The average MI classification accuracies of recently developed algorithms CTWS [26] and LRFCSP [20] over seven subjects are compared with VaS-LDA and VaS-SVM$_{lk}$ as illustrated in Table 4.

It is observed that the average classification accuracies of both VaS-LDA and VaS-SVM$_{lk}$ are higher than that of LRFCSP [20], CTWS [26] as well as CSP-SVM over the seven subjects. The highest average accuracy (89.84%) is achieved by VaS-LDA compared to all the methods mentioned in Table 4. The maximum accuracies only for subjects 'g' (93.20%) and 'e' (99.00%) are attained by the methods LRFCSP and CTWS respectively. The proposed method VaS-LDA exhibits maximum accuracy for three subjects 'c' (85.60%), 'd' (97.60%) and 'f' (90.80%), whereas, VaS-SVM$_{lk}$ attains highest classification accuracy with two subjects 'a' (92.50%) and 'b' (77.00%). Although VaS-SVM$_{lk}$ performs better than VaS-LDA over four subjects (illustrated in Table 3), the maximum average accuracy is achieved by VaS-LDA for seven subjects (presented in Table 4). The average classification accuracy (over seven subjects) of VaS-LDA is 0.24%, 5.14% and 4.98% higher than that of VaS-SVM$_{lk}$, LRFCSP [20] and CTWS [26] respectively. The results imply the superiority of the proposed method VaS-LDA over the seven subjects of Dataset I. The statistical test is performed to check the significance of the proposed method. Tukey-Kramer based post-hoc test [45] suggests that the proposed VaS-LDA achieves significantly higher accuracy across the seven subjects for motor imagery classification than the other methods (VaS-LDA vs VaS-SVM$_{lk}$: $p>0.12$; VaS-LDA vs CSP-SVM: $p<0.05$; VaS-LDA vs CTWS: $p>0.06$; VaS-LDA vs LRFCSP: $p>0.07$).

TABLE 4. Comparison of performance of VaS-LDA and VaS-SVM$_{lk}$ with recently reported methods in term of classification accuracy (in %) for dataset I.

Subject	LRFCSP [20]	CTWS [26]	CSP-SVM	VaS-LDA	VaS-SVM$_{lk}$
a	87.40	83.00	88.70	88.40	**92.50**
b	70.00	67.00	73.50	75.60	**77.00**
c	67.40	85.50	78.70	**85.60**	82.70
d	92.90	93.00	94.60	**97.60**	96.40
e	93.40	**99.00**	94.10	98.50	97.20
f	88.80	85.50	87.90	**90.80**	88.80
g	**93.20**	81.00	92.20	92.40	92.60
Mean	84.70	84.86	87.10	**89.84**	89.60
SD	5.78	10.4	7.46	7.80	**7.41**

TABLE 5. Comparison of performance (in term of classification accuracy) among different recent methods evaluated with dataset II.

Reference	Method	Mean accuracy (%)	SD
M.-H. Lee et.al. [42]	CSP with 10-fold cross validation	72.20	15.40
O-Y. Kwon et.al. [19]	Subject-dependent	71.20	15.88
Proposed method	CSP-SVM	70.40	16.30
	VaS-LDA	**73.85**	15.25
	VaS-SVM$_{lk}$	73.50	**14.80**

The performance of the proposed methods is also evaluated with Dataset II. All of 54 subjects [19] of Dataset II are considered to conduct the experiments and the results are compared with the recently developed algorithms as illustrated in Table 5. The average results as well as standard deviations over the subjects are illustrated to perform the comparison.

It is observed in Table 5 that the performances (in term of classification accuracy) of the proposed methods (VaS-LDA and VaS-SVM$_{lk}$) are higher than that of the recently developed algorithms described in [19], [42]. In [42], a number of methods are studied using Dataset II and the maximum

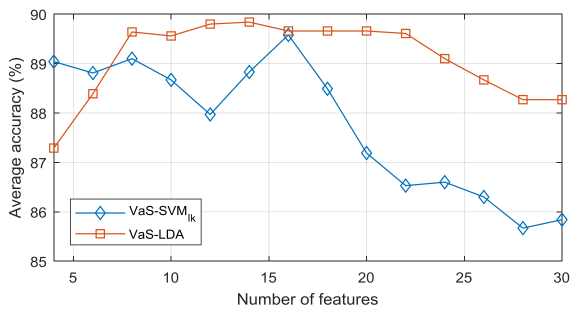

FIGURE 6. Performance of VaS-LDA and VaS-SVM $_{lk}$ in term of average classification accuracy (%) over the seven subjects of dataset I as a function of the number of features.

accuracy (72.20%) is achieved using CSP with 10-fold cross validation for the data recorded in Session 2. This study is subject-dependent and hence the performance is compared only with subject-dependent case illustrated in [19]. The performances of the proposed methods (VaS-LDA and VaS-SVM$_{lk}$) are higher than that of the recently reported algorithms [19], [42] as well as the baseline method (CSP-SVM). The session 2 of Dataset II is used in all cases to evaluate the performances.

The decomposition of individual channel into narrowband signals localizes the discriminative components yielding potential features to improve the classification accuracy. The individual trail is regenerated using the subbands obtained by multivariate wavelet transform. Two different subband arrangements are implemented. The CSP is applied to the narrowband signal space to derive discriminative features. The number of features selected to be used in classification is an important factor affecting the classification accuracy. The average accuracies for VaS-LDA and VaS-SVM$_{lk}$ methods as a function of the number of features are illustrated in Fig. 6.

It is observed that the maximum accuracy is achieved with 14 and 16 features for VaS-LDA and VaS-SVM$_{lk}$ methods respectively. The performance of SVM based method is decreasing with increased number of features. The use of higher number of features certainly includes irrelevant features. Too many features may render the convergence difficult, leading to random classification decisions. Hence, the accuracy is decreasing with higher number of features in SVM. The higher number of features also requires high computational complexity. Considering the tradeoff between computational complexity and performance, the 14 features are selected with VaS-LDA method as an effective motor imagery classification approach using EEG.

VI. CONCLUSION

A novel method is introduced for motor imagery classification in BCI paradigm using trial regeneration approach. The motor imagery task is captured by multichannel EEG within a short time trial. The individual channel contains a wideband signal. The narrowband signal contains more discriminative features for event classification. Each channel is decomposed into four subbands using multivariate wavelet transform. The lowest frequency subband contains the signal components (0 – 6 Hz) out of the target frequency range 8 – 30 Hz and hence it is discarded. Other three subbands are arranged in two ways (VaS and HaS) as described in Section II (B) to regenerate the trials with narrowband signals. The features are extracted from each of arrangement using well known CSP method. Two classifiers – LDA and SVM are used to classify the motor imagery tasks. Thus four different combinations namely VaS-LDA, HaS-LDA, VaS-SVM and HaS-SVM are evaluated in term of classification accuracy. Also the performance of SVM classifier is evaluated for different kernel functions. It is found that SVM performs better with linear and radial basis function (RBF) kernels for VaS and HaS arrangement respectively. The maximum average performance over the subjects is achieved by using VaS-LDA for both datasets. The proposed method is compared with recently developed algorithms and it is observed that the VaS-LDA performs better than others. The effect of the number of features is also studied to select the optimal feature dimension considering the computation cost and classification performance. The future extension of this study is to implement the proposed method in multiclass BCI paradigm with discriminative feature selection approach.

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(Md. Khademul Islam Molla and Sanjoy Kumar Saha contributed equally to this work.)

REFERENCES

[1] J. Wolpaw, N. Birbaumer, D. McFarland, G. Pfurtscheller, and T. Vaughan, "Brain-computer interfaces for communication and control," *Clin. Neurophys.*, vol. 113, no. 6, pp. 767–791, Jun. 2002.

[2] X. Wan, K. Zhang, S. Ramkumar, J. Deny, G. Emayavaramban, M. S. Ramkumar, and A. F. Hussein, "A review on electroencephalogram based brain computer interface for elderly disabled," *IEEE Access*, vol. 7, pp. 36380–36387, 2019.

[3] Y. Wang, X. Gao, B. Hong, C. Jia, and S. Gao, "Brain-computer interfaces based on visual evoked potentials," *IEEE Eng. Med. Biol. Mag.*, vol. 27, no. 5, pp. 64–71, Sep. 2008.

[4] T. Kayikcioglu and O. Aydemir, "A polynomial fitting and k-NN based approach for improving classification of motor imagery BCI data," *Pattern Recognit. Lett.*, vol. 31, no. 11, pp. 1207–1215, Aug. 2010.

[5] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, "Optimizing spatial filters for robust EEG single-trial analysis," *IEEE Signal Process. Mag.*, vol. 25, no. 1, pp. 41–56, Dec. 2008.

[6] G. Pfurtscheller, C. Brunner, A. Schlögl, and F. H. L. da Silva, "Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks," *NeuroImage*, vol. 31, no. 1, pp. 153–159, May 2006.

[7] W. Wu, X. Gao, B. Hong, and S. Gao, "Classifying single-trial EEG during motor imagery by iterative spatio-spectral patterns learning (ISSPL)," *IEEE Trans. Biomed. Eng.*, vol. 55, no. 6, pp. 1733–1743, Jun. 2008.

[8] T. Yu, J. Xiao, F. Wang, R. Zhang, Z. Gu, A. Cichocki, and Y. Li, "Enhanced motor imagery training using a hybrid BCI with feedback," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 7, pp. 1706–1717, Jul. 2015.

[9] G. Pfurtscheller, "Mapping of event-related desynchronization and type of derivation," *Electroencephalogr. Clin. Neurophysiol.*, vol. 70, no. 2, pp. 190–193, Aug. 1988.

[10] D. J. McFarland, L. M. McCane, S. V. David, and J. R. Wolpaw, "Spatial filter selection for EEG-based communication," *Electroencephalogr. Clin. Neurophysiol.*, vol. 103, no. 3, pp. 386–394, Sep. 1997.

[11] C. Park, D. Looney, N. U. Rehman, A. Ahrabian, and D. P. Mandic, ''Classification of motor imagery BCI using multivariate empirical mode decomposition,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 21, no. 1, pp. 10–22, Jan. 2013.
[12] S. Siuly and Y. Li, ''Improving the separability of motor imagery EEG signals using a cross correlation-based least square support vector machine for Brain–Computer interface,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 20, no. 4, pp. 526–538, Jul. 2012.
[13] M. R. Islam, T. Tanaka, and M. K. I. Molla, ''Multiband tangent space mapping and feature selection for classification of EEG during motor imagery,'' *J. Neural Eng.*, vol. 15, no. 4, Jun. 2018, Art. no. 046021.
[14] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, ''Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b,'' *Frontiers Neurosci.*, vol. 6, p. 39, Mar. 2012.
[15] S. Lemm, B. Blankertz, G. Curio, and K.-R. Müller, ''Spatio-spectral filters for improving the classification of single trial EEG,'' *IEEE Trans. Biomed. Eng.*, vol. 52, no. 9, pp. 1541–1548, Sep. 2005.
[16] M. K. I. Molla, A. A. Shiam, M. R. Islam, and T. Tanaka, ''Discriminative feature selection-based motor imagery classification using EEG signal,'' *IEEE Access*, vol. 8, pp. 98255–98265, 2020.
[17] J. Jin, R. Xiao, I. Daly, Y. Miao, X. Wang, and A. Cichocki, ''Internal feature selection method of CSP based on L1-norm and dempster-shafer theory,'' *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 24, 2020, doi: 10.1109/TNNLS.2020.3015505.
[18] H. Wang, T. Xu, C. Tang, H. Yue, C. Chen, L. Xu, Z. Pei, J. Dong, A. Bezerianos, and J. Li, ''Diverse feature blend based on filter-bank common spatial pattern and brain functional connectivity for multiple motor imagery detection,'' *IEEE Access*, vol. 8, pp. 155590–155601, 2020.
[19] O.-Y. Kwon, M.-H. Lee, C. Guan, and S.-W. Lee, ''Subject-independent brain–computer interfaces based on deep convolutional neural networks,'' *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 10, pp. 3839–3852, Oct. 2020.
[20] Y. Park and W. Chung, ''Frequency-optimized local region common spatial pattern approach for motor imagery classification,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 7, pp. 1378–1388, Jul. 2019.
[21] J. Jin, Y. Miao, I. Daly, C. Zuo, D. Hu, and A. Cichocki, ''Correlation-based channel selection and regularized feature optimization for MI-based BCI,'' *Neural Netw.*, vol. 118, pp. 262–270, Oct. 2019.
[22] H. Chang and J. Yang, ''Automated selection of a channel subset based on the genetic algorithm in a motor imagery brain-computer interface system,'' *IEEE Access*, vol. 7, pp. 154180–154191, 2019.
[23] Y. Park and W. Chung, ''Optimal channel selection using correlation coefficient for CSP based EEG classification,'' *IEEE Access*, vol. 8, pp. 111514–111521, 2020.
[24] Y. Park and W. Chung, ''Selective feature generation method based on time domain parameters and correlation coefficients for filter-bank-CSP BCI systems,'' *Sensors*, vol. 19, no. 17, p. 3769, Aug. 2019.
[25] J. Jin, C. Liu, I. Daly, Y. Miao, S. Li, X. Wang, and A. Cichocki, ''Bispectrum-based channel selection for motor imagery based brain-computer interfacing,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 10, pp. 2153–2163, Oct. 2020.
[26] J. Feng, E. Yin, J. Jin, R. Saab, I. Daly, X. Wang, D. Hu, and A. Cichocki, ''Towards correlation-based time window selection method for motor imagery BCIs,'' *Neural Netw.*, vol. 102, pp. 87–95, Jun. 2018.
[27] N. E. Huang, Z. Shen, S. R. Long, M. C. Wu, H. H. Shih, Q. Zheng, N.-C. Yen, C. C. Tung, and H. H. Liu, ''The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis,'' *Proc. Roy. Soc. London A, Math., Phys. Eng. Sci.*, vol. 454, no. 1971, pp. 903–995, Mar. 1998.
[28] M. K. I. Molla, M. R. Islam, T. Tanaka, and T. M. Rutkowski, ''Artifact suppression from EEG signals using data adaptive time domain filtering,'' *Neurocomputing*, vol. 97, pp. 297–308, Nov. 2012.
[29] C. Park, D. Looney, P. Kidmose, M. Ungstrup, and D. P. Mandic, ''Time-frequency analysis of EEG asymmetry using bivariate empirical mode decomposition,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 19, no. 4, pp. 366–373, Aug. 2011.
[30] C. M. Sweeney-Reed and S. J. Nasuto, ''A novel approach to the detection of synchronisation in EEG based on empirical mode decomposition,'' *J. Comput. Neurosci.*, vol. 23, no. 1, pp. 79–111, Jun. 2007.
[31] N. U. Rehman and D. P. Mandic, ''Filter bank property of multivariate empirical mode decomposition,'' *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2421–2426, May 2011.
[32] R. Bracewell, *The Fourier Transform and Its Applications*. New York, NY, USA: McGraw-Hill, Feb. 1986.
[33] I. Daubechies, J. Lu, and H.-T. Wu, ''Synchrosqueezed wavelet transforms: An empirical mode decomposition-like tool,'' *Appl. Comput. Harmon. Anal.*, vol. 30, no. 2, pp. 243–261, Mar. 2011.
[34] Q. Zhang and A. Benveniste, ''Wavelet networks,'' *IEEE Trans. Neural Netw.*, vol. 3, no. 6, pp. 889–898, Nov. 1992.
[35] K. G. Oweiss and D. J. Anderson, ''Noise reduction in multichannel neural recordings using a new array wavelet denoising algorithm,'' *Neurocomputing*, vols. 38–40, pp. 1687–1693, Jun. 2001.
[36] B. Hosseinifard, M. H. Moradi, and R. Rostami, ''Classifying depression patients and normal subjects using machine learning techniques and nonlinear features from EEG signal,'' *Comput. Methods Programs Biomed.*, vol. 109, no. 3, pp. 339–345, Mar. 2013.
[37] Z. Zhang, F. Duan, J. Sole-Casals, J. Dinares-Ferran, A. Cichocki, Z. Yang, and Z. Sun, "A novel deep learning approach with data augmentation to classify motor imagery signals," *IEEE Access*, vol. 7, pp. 15945–15954, 2019.
[38] C.-Y. Yeh, W.-P. Su, and S.-J. Lee, ''An efficient multiple-kernel learning for pattern classification,'' *Expert Syst. Appl.*, vol. 40, no. 9, pp. 3491–3499, Jul. 2013.
[39] C. Neuper, G. Müller-Putz, R. Scherer, and G. Pfurtscheller, ''Motor imagery and EEG-based control of spelling devices and neuroprostheses,'' *Prog. Brain Res.*, vol. 159, no. 10, pp. 393–409, Jan. 2006.
[40] G. H. Klem, H. O. Lüeders, H. H. Jasper, and C. Elger, ''The ten-twenty electrode system of the international federation,'' *Electroencephalogr. Clin. Neurophysiol.*, vol. 52, no. 3, pp. 3–6, 1999.
[41] X. Bai, X. Wang, S. Zheng, and M. Yu, ''The offline feature extraction of four-class motor imagery EEG based on ICA and wavelet-CSP,'' in *Proc. 33rd Chin. Control Conf.*, Jul. 2014, pp. 7189–7194.
[42] M. Lee, O. Kwon, Y. Kim, H. Kim, Y. Lee, J. Williamson, S. Fazli, and S. Lee, ''EEG dataset and OpenBMI toolbox for three BCI paradigms: An investigation into BCI illiteracy,'' *GigaScience*, vol. 8, no. 5, May 2019, Art. no. giz002.
[43] N. U. Rehman and D. P. Mandic, ''Multivariate empirical mode decomposition,'' *Proc. R. Soc. Lond. A, Math., Phys. Eng. Sci.*, vol. 466, no. 2117, pp. 1291–1302, Dec. 2009.
[44] S. Mallat, *A Wavelet Tour of Signal Processing*. Amsterdam, The Netherlands: Elsevier, Sep. 2009.
[45] M. K. I. Molla, N. Morikawa, M. R. Islam, and T. Tanaka, ''Data-adaptive spatiotemporal ERP cleaning for single-trial BCI implementation,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 7, pp. 1334–1344, Jul. 2018.
[46] K.-C. Hsu and S.-N. Yu, ''Detection of seizures in EEG using subband nonlinear parameters and genetic algorithm,'' *Comput. Biol. Med.*, vol. 40, no. 10, pp. 823–830, Oct. 2010.
[47] M. K. I. Molla, T. Tanaka, T. Osa, and M. R. Islam, ''EEG signal enhancement using multivariate wavelet transform application to single-trial classification of event-related potentials,'' in *Proc. IEEE Int. Conf. Digit. Signal Process.*, Jul. 2015, pp. 804–808.
[48] Y. Zhang, G. Zhou, J. Jin, X. Wang, and A. Cichocki, ''Optimizing spatial patterns with sparse filter bands for motor-imagery based brain–computer interface,'' *J. Neurosci. Methods*, vol. 255, pp. 85–91, Nov. 2015.
[49] A. Subasi and M. I. Gursoy, ''EEG signal classification using PCA, ICA, LDA and support vector machines,'' *Expert Syst. Appl.*, vol. 37, no. 12, pp. 8659–8666, Dec. 2010.
[50] R. Duda and P. Hart, *DG Stork Pattern Classification*. Hoboken, NJ, USA: Wiley, 2001.
[51] K. Fukunaga, *Introduction to Statistical Pattern Recognition*. Amsterdam, The Netherlands: Elsevier, Oct. 2013.
[52] C. Cortes and V. Vapnik, ''Support-vector networks,'' *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.
[53] X. Chen, J. Yang, Q. Ye, and J. Liang, ''Recursive projection twin support vector machine via within-class variance minimization,'' *Pattern Recognit.*, vol. 44, nos. 10–11, pp. 2643–2655, Oct. 2011.
[54] O. Mangasarian and D. Musicant, ''Lagrangian support vector machines,'' *J. Mach. Learn. Res.*, vol. 1, pp. 161–177, Mar. 2001.

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